Fundamental Concepts in General Topology

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Our text is a short review of the core notations of general topology, often purely called *point-set topology*. This mathematical theory deals with the study of limits, continuity and geometric properties of the underlying sets. Rather topology has meaning in its own right, it mainly serves as foundation of further mathematical areas like *functional analysis*, *differential topology*, *Riemannian geometry* and other ones. Topological concepts are frequently required to understand mathematical analysis on a high and abstract level. One may justly claim that analysis today is unthinkable without including the extremely useful notions provided by topology.

The content is divided into four parts: first of all, we outline the basic concepts of topology. In the subsequent parts we shall have a look at specific kinds of topological spaces, which especially are *metric spaces* and *topological groups*.

I. Topological Spaces

This first part deals with the main concepts of topology, and the first one has to consider is that of a *topological space*.

Given a non-empty set X, any collection $\mathcal{T}$ of subsets of X is called a *topology* on X iff the following conditions are fulfilled:

(T-1) both, the empty set $\emptyset$ and the entire set X are elements of $\mathcal{T}$,

(T-2) if $\mathcal{O}_i \in \mathcal{T}$ for $i = 1, \ldots, n$, then $\mathcal{O}_1 \cap \ldots \cap \mathcal{O}_n \in \mathcal{T}$.

(T-3) if $\{\mathcal{O}_\alpha\}$ is any collection of members of $\mathcal{T}$, then $\bigcup \mathcal{O}_\alpha \in \mathcal{T}$.

This means that a topology is closed under the formation of finitely many intersections and arbitrarily many unions. A *topological space* is a pair $(X, \mathcal{T})$ consisting of a set X and a topology $\mathcal{T}$. It is convenient to denote the elements of a topological space as *points*.

1. Open and Closed Sets. For any topological space $(X, \mathcal{T})$ the members of the topology $\mathcal{T}$ are called the *open sets* in X. On the other hand, any subset of X is said to be *closed*, if its complementary set is open. Due to the properties of an open set, the intersection of arbitrarily many closed sets as well as the union of a *finite* number of closed sets is closed again. Especially the base set X and the

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empty set are both closed.

The framework of a topological space also enables the definition of some further important concepts in point-set topology:

(a) If $\mathcal{A}$ is a subset of a topological space X, then the set $\mathring{\mathcal{A}}$ denotes the union of all open subsets of $\mathcal{A}$, or equivalently, it is the largest open set $\mathcal{O}$ such that $\mathcal{O} \subset \mathcal{A}$.

(b) The set $\mathring{\mathcal{A}}$ is said to be the *interior* of $\mathcal{A}$, and each element of $\mathring{\mathcal{A}}$ is called an *inner point* of $\mathcal{A}$.

(c) Conversely, consider for a given set $\mathcal{A}$ the intersection $\overline{\mathcal{A}}$ of all closed sets containing $\mathcal{A}$. Then the set $\overline{\mathcal{A}}$ is closed again and is called the *closure* of $\mathcal{A}$, and each point $x \in \overline{\mathcal{A}}$ is said to be a *limit point* of $\mathcal{A}$.

In summary, for each subset $\mathcal{A} \subset X$ there is always the sequence of inclusions

$$\emptyset \subseteq \mathring{\mathcal{A}} \subseteq \mathcal{A} \subseteq \overline{\mathcal{A}} \subseteq X. \tag{1.1}$$

We take care of the midth subchain $\mathring{\mathcal{A}} \subseteq \mathcal{A} \subseteq \overline{\mathcal{A}}$. If the inclusions can be replaced by equality, that is, $\mathring{\mathcal{A}} = \mathcal{A}$ on the left side, then and only then the set $\mathcal{A}$ is open. Similarly, if $\mathcal{A} = \overline{\mathcal{A}}$ on the right side, then and only then the set $\mathcal{A}$ is closed.

Now consider the first inclusion $\emptyset \subseteq \mathring{\mathcal{A}}$ in (1.1). If it can be replaced by the equality $\emptyset = \mathring{\mathcal{A}}$, then $\mathcal{A}$ is called *discrete*. On the other hand, in case of $\overline{\mathcal{A}} = X$, the set $\mathcal{A}$ is called *dense* in X, that is, every point $x \in X$ can be obtained as a limit point of $\mathcal{A}$.

Finally, for a given subset $\mathcal{A} \subset X$, the relative complement set $\partial\mathcal{A} := \overline{\mathcal{A}} \setminus \mathring{\mathcal{A}}$ is called the *boundary* of $\mathcal{A}$, and each element of $\partial\mathcal{A}$ is said to be a *boundary point* of $\mathcal{A}$. It can be shown that a set $\mathcal{O} \subset X$ is open iff it contains no boundary points, and a set $\mathcal{A} \subset X$ is closed iff it contains all its boundary points. We obtain the boundary of $\mathcal{A}$ as the intersection of the sets $\overline{\mathcal{A}}$ and $\overline{X \setminus \mathcal{A}}$, which implies that a boundary set is always closed.

2. Neighbourhoods. The notion of *neighbourhood* is one of the most important one in general topology. Roughly spoken, any set constitutes a neighbourhood of a point $x \in X$, if it contains all points sufficiently close to x. The exact definition states as follows: a subset $\mathcal{U} \subset X$ is called a *neighbourhood* of $x \in X$, iff x is an interior point of $\mathcal{U}$. For example, an open set is a neighbourhood for all of its elements. The neighbourhoods of a given point $x \in X$ have the property, that each finite intersection and each union of

arbitrarily many neighbourhoods of x is also a neighbourhood of x. Moreover, every superset $\mathcal{U}' \supset \mathcal{U}$ of a neighbourhood of x forms a neighbourhood of x.

Neighbourhoods are useful to define the concept of *convergence* for sequences in a topological space X. In particular, a sequence (ϕ_n) is called *convergent* to an element $x \in X$, if for each neighbourhood $\mathcal{U}$ of x there are only finitely many members of (ϕ_n) being outside $\mathcal{U}$. We will study the convergence of sequences again in the setting of metric spaces (see Part II).

A *neighbourhood base* of a point x is a family $\mathfrak{F}$ of neighbourhoods of x with the additional property that each open set $\mathcal{O}_x$ with $x \in \mathcal{O}_x$ contains a neighbourhood $\mathcal{U}_x \in \mathfrak{F}$ as a subset: $\mathcal{U}_x \subset \mathcal{O}_x$. Furthermore, a topological space is said to satisfy the *first countability axiom*, if each point has a mostly *countable* neighbourhood base.

3. Hausdorff's Separation Axiom. Every set X containing at least two points can be employed with different topologies. Obviously, the largest one is the *discrete* topology on X, which coincides with the power set of X. On the other hand, the coarsest thinkable topology on X consists of the empty set and the set X only.

If $\mathcal{T}_1, \mathcal{T}_2$ are topologies on X, then $\mathcal{T}_1$ is called *weaker* than $\mathcal{T}_2$, if every open set in $\mathcal{T}_1$ is also an open set in $\mathcal{T}_2$, and at the same time $\mathcal{T}_2$ is called *stronger* than $\mathcal{T}_1$. The notion of topology on a set X is extremely general, and it includes pathological examples of very little interest such as the topology whose only elements are the sets $\emptyset$ and X.

For most applications in analysis, it is desirable to impose additional conditions to the topology that guarantee the existence of sufficiently many open sets. The most important condition of this kind is the *Hausdorff separation axiom*, which can be stated as follows: a topological space X is called *Hausdorff space*, if for any points $x, y \in X$ there are open sets $\mathcal{O}_x$ and $\mathcal{O}_y$ such that $x \in \mathcal{O}_x$, $y \in \mathcal{O}_y$ and $\mathcal{O}_x \cap \mathcal{O}_y = \emptyset$. In other words, in a Hausdorff space every pair of points can be separated by suitable open sets with empty intersection. Besides many other separation axioms in topology the Hausdorff one is the most important of them, because nearly all important topological spaces considered in analysis are Hausdorff spaces. The Hausdorff separation axiom ensures two desirable properties of a topological space: (1) each singleton $\{x\}$ is closed, and (2) the limit point of any convergent sequence is unique.

4. Continuous Mappings. The notion of *continuity* of a real-valued function at a certain point $t \in \mathbb{R}$ is well known from elementary calculus. Using neighbourhoods, it is possible to extend this notion to arbitrary mappings between topological spaces X, Y as follows: a mapping $f : X \to Y$ is said to be *continuous at a point* $x \in X$, if for each neighbourhood $\mathcal{V}$ of $f(x)$ there is a neighbourhood $\mathcal{U}$ of x with $f(\mathcal{U}) \subseteq \mathcal{V}$. Furthermore, the mapping f is called *continuous*, iff f is continuous at *every* point $x \in X$.

By means of open sets, an equivalent definition of continuity is the following one, which is the most common definition in general topology: a mapping $f : X \to Y$ is called *continuous*, if the preimages $f^{-1}(\mathcal{O})$ of open sets $\mathcal{O} \subseteq Y$ are also open sets in X. Obviously, the property of a mapping being continuous depends on the choice of the topology in both spaces.

If a continuous mapping $f : X \to Y$ is bijective, we cannot conclude that the inverse mapping $f^{-1} : Y \to X$ is continuous, too. This fact justifies the notion of *homeomorphism* between topological spaces: a bijective continuous mapping $f : X \to Y$ is called a *homeomorphism*, if the inverse mapping f^{-1} is continuous as well. Thus a homeomorphism carries over exactly the open sets in open sets again. If there is a homeomorphism between the topological spaces X and Y, then the spaces are called *homeomorphic* to each other (denoted $X \cong Y$). In this case we can talk about structural identity of the spaces X and Y with respect to their topological properties. A property of a topological space is called a *topological invariant*, if it is preserved by homeomorphic mappings.

5. Connected Sets. Open sets are suitable to define the notion of *connectedness* for subsets of a topological space: a set $C \subseteq X$ is said to be *connected*, if there are *not* two open sets $\mathcal{O}_1, \mathcal{O}_2$ with $\mathcal{O}_1 \cap \mathcal{O}_2 = \emptyset$ and $C \subseteq \mathcal{O}_1 \sqcup \mathcal{O}_2$, otherwise it is called *disconnected*. Each open and connected set is usually called a *domain*.

Clearly, the property of a set $C \subset X$ being connected depends on the topology of the underlying space. Connected sets mapped by a continuous function do not lose this property. More precisely, if $\mathcal{A} \subset X$ is connected and the mapping $f : X \to Y$ is continuous, then $f(\mathcal{A}) \subseteq Y$ is connected as well. Connectedness is actually a topological invariant.

The most well-known criterion for connectedness is given by the following theorem: *a topological space X is connected, iff X and the*

empty set $\emptyset$ are the only sets, which are open and closed together.

For a topological space X, a *connected component* of a point $x \in X$ is defined as the union of all connected sets which contain this point. Each such set is obviously connected and cannot be extended to a larger connected set by adding further points. Since the closure of a connected set is connected again, the connected components of a topological space are always closed. They form a disjoint decomposition of the space into connected subsets, being the coarsest qualitative information about the global structure of a space.

6. Generators of New Topologies. It can be shown that for a base set X the intersection of arbitrarily many topologies on X forms a topology on X again. Thus for a given collection $\mathcal{S}$ of subsets of X, we define the topology $[\mathcal{S}]$ to be the intersection of all topologies on X containing $\mathcal{S}$. The set $\mathcal{S}$ is denoted a *generator* of the topology $[\mathcal{S}]$, and reversely $[\mathcal{S}]$ is said to be *generated* by $\mathcal{S}$.

As a first application of this method we define the *standard topology* $\mathcal{T}_{\mathbb{R}}$ on the real line $\mathbb{R}$ as the topology generated by all half-bounded open intervals, that is, $\mathcal{T}_{\mathbb{R}}$ is the smallest topology on $\mathbb{R}$ containing the half-bounded intervals $(-\infty, a)$ and (b, ∞).

Let X be any set and let $f_i : X \to Y_i$ $(i \in \mathcal{I})$ be a family of mappings into arbitrarily many topological spaces Y_i. We consider the collection $\mathcal{S}$ of preimages $f_i^{-1}(\mathcal{O})$ for all the open sets $\mathcal{O} \subset Y_i$. Then the topology $[\mathcal{S}]$ generated by $\mathcal{S}$ is the coarsest topology on X such that all mappings f_i remain still continuous. It is called the *initial topology* with respect to the family $\{f_i\}_{i \in \mathcal{I}}$.

Using initial topologies, we introduce the *subset topology* and the *product topology*:

1. Let X be a topological space and $M \subset X$ be a proper subset of X. We define the *subset topology* $\mathcal{T}_M$ on M as the initial topology with respect to the inclusion mapping $M \hookrightarrow X$, that is, the subset topology is the smallest one such that the inclusion of M into X is continuous. The pair $(M, \mathcal{T}_M)$ is a topological space in its own right, and a subset $\mathcal{O} \subset M$ is open with respect to $\mathcal{T}_M$ iff there is an open set $\mathcal{O}'$ in X such that $\mathcal{O} = M \cap \mathcal{O}'$.

   It should be mentioned that the subset topology in M can also be received by definition of its open sets,

$$\mathcal{T}_M := \{M \cap \mathcal{O}' : \mathcal{O}' \in \mathcal{T}\}.$$

A *continuous embedding* of a space X into a space Y is a mapping $\iota : X \to Y$ such that the images of open sets in X under ι exactly form the subset topology of $\iota(X)$ considered as a subspace of Y.

2. The *product topology* $\mathcal{T}_{X\times Y}$ on the cartesian product $X \times Y$ is defined to be the initial topology with respect to the projection mappings $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$. In other words, it is the coarsest topology for which the both mappings are continuous. Note that this definition of the product topology can be extended to the cartesian product of arbitrarily many topological spaces. Another way to define the product topology on the set $X \times Y$ is given by the following one: $\mathcal{T}_{X\times Y}$ is the topology generated by the set of all pairs $(\mathcal{O}_1, \mathcal{O}_2)$ with $\mathcal{O}_1 \in X$ and $\mathcal{O}_2 \in Y$.

We can use the product topology to define the *standard topology* $\mathcal{T}_{\mathbb{C}}$ on the field $\mathbb{C}$ of complex numbers, since $\mathbb{C}$ can be considered as the cartesian product $\mathbb{R} \times \mathbb{R}$. Here, for each factor of $\mathbb{R} \times \mathbb{R}$ the standard topology of the space $\mathbb{R}$ is used.

7. Compact and Locally Compact Spaces. One of the most useful concepts coming from topology is that of *compactness*. It is a property of finiteness defined for subsets of a topological space. Preliminary, given a topological space X, an *open covering* of a set $K \subseteq X$ is a collection $\mathfrak{F}$ of open sets such that

$$K \subseteq \bigcup_{\mathcal{O}\in\mathfrak{F}} \mathcal{O}.$$

The set K is called *compact*, if *every open covering of K has a finite subcovering*.

The meaning of compactness is, that many proofs are much easier and more transparent, if the set under consideration is assumed to be compact. In general, when examining a topological problem it is a benefite to consider primarily the compact case if possible. Note that compactness is an inner property of a subset, that is, a subset $K \subset X$ is compact with respect to the topology $\mathcal{T}$ on X if and only if it is compact in its subset topology $\mathcal{T}_K$.

The property of compactness of a set is inherited to its subsets supposed these are closed, that is, every closed subset of a compact set or space is compact, too. On the other hand, if a topological space X is Hausdorff, then every compact subset of X is closed. We shall call a subset *relatively compact*, if its closure is compact.

Compactness provides also a helpful criterion to decide whether a given map $f: X \to Y$ is homeomorphic. In particular, let X be a compact space and Y be a Hausdorff space, then each continuous and bijective mapping is a homeomorphism.

Since compactness is a rather ambitious property, topological spaces having this property are very sparse. But there is a weaker property, called *locally compactness*, which is fulfilled by a larger number of topological spaces. Locally compactness is a property which has to be satisfied by all the elements of a space: a topological space X is called *locally compact*, if each neighbourhood of any point $x \in X$ contains a compact neighbourhood. In case of X being a Hausdorff space, X is locally compact if and only if each point $x \in X$ has a compact neighbourhood.

The class of *locally compact spaces* plays an outstanding role for investigations in other mathematical areas like *measure theory* and *abstract harmonic analysis*. Nearly all locally compact spaces of practical interest are Hausdorff spaces as well, so the notion *LCH space* is a popular abbreviation for a topological space being both, namely *locally compact* and *Hausdorff*.

8. Compact-Open Topologies. Given the two sets X and Y, we denote Y^X the set of all mappings $f: X \to Y$. In additon, if both sets are topological spaces, let $C(X,Y) \subset Y^X$ be the subset of *continuous* mappings $f: X \to Y$. It is our goal to equip the set $C(X,Y)$ with a suitable topology.

Preliminary, for a compact subset $K \subset X$ and an open subset $U \subset Y$ we introduce the family of mappings

$$S(K,\mathcal{O}) := \{f \in C(X,Y) : f(K) \subset \mathcal{O}\}.$$

Then the set $C(X,Y)$ can be furnished with the topology $\mathcal{T}_{co}$, which is generated by the collection of all such sets $S(K,\mathcal{O})$, called the *compact-open topology* on $C(X,Y)$. It turns out that the set $C(X,Y)$ equipped with this topology is a Hausdorff space whenever Y is Hausdorff.

The establishment of the compact-open topology on sets of continuous mappings can be seen as a standard method in general topology. Its meaning is based on the fact that functions in $C(X,Y)$ are closed together, if their images on compact subsets are so.

In case of $Y = \mathbb{R}$ or $Y = \mathbb{C}$, let both sets be endowed with their standard topology. Then for the spaces $C(X, \mathbb{R})$ and $C(X, \mathbb{C})$ we get the compact-open topology, respectively. Since $\mathbb{R}$ and $\mathbb{C}$ are Hausdorff spaces according to the standard topologies, the functions sets $C(X, \mathbb{R})$ and $C(X, \mathbb{C})$ prove to be Hausdorff spaces, too.

9. Baire Spaces. A topological space is called a *Baire space* (due to R.L. BAIRE), if the union of a countable collection of closed sets with empty interior has empty interior, too. This definition is equivalent to the following one: if the union of countable many closed subsets has an interior point, then at least one of them must have an interior point. BAIRE*'s category theorem* claims (as a special variant of a more general theorem) that *every LCH space is a Baire space.* The word "category" is due to BAIRE's original terminology, according to which a set is said to be *of first category* if it is a countable union of nowhere dense sets, and *of second category* otherwise. We may reformule BAIRE's theorem by quoting that *all LCH spaces are of second category.*

II. Metric Spaces

The theory of *metric spaces* may serve as a link between general topology and modern analysis. Introduced by M. FRÉCHET, they are one of the most important classes within that of topological spaces. Roughly spoken, they can be considered as a straightforward generalization of the Euclidean spaces. The main property of a metric space is that a distance value, called the *metric* on that space, is assigned to each pair of its elements. This leads to an intuitive understanding of the topological concept of *closeness* between points.

1. Metrics. Given a non-empty set M, a mapping $d : M \times M \to \mathbb{R}$ is called a *metric* on M iff the following four conditions hold for all $x, y, z \in M$:

(M-1) $d(x, y) \geq 0$,
(M-2) $d(x, y) = 0$ if and only if $x = y$,
(M-3) $d(x, y) = d(y, x)$;
(M-4) $d(x, z) \leq d(x, y) + d(y, z)$ (*triangle inequality*).

If d is a metric on the set M, then the pair (M, d) is called a *metric space.* Any set M can be made into a metric space in different ways by employing different metric functions.

There is a natural way to furnish a metric space with a distinguished topology. Let (M, d) be a metric space, then any set of the form $B_\epsilon(x) := \{y \in M : d(x,y) < \epsilon\}$ is called an *ϵ-ball* around a point $x \in M$. We then introduce open sets in M by means of ϵ-balls: a subset $\mathcal{O} \subseteq M$ is called open, if for every point $x \in \mathcal{O}$ there is a ball $B_\epsilon(x)$ such that $B_\epsilon(x) \subset \mathcal{O}$.

Indeed, the system $\mathcal{T}_d$ of all such open sets in M forms a topology and is called the *induced* or *metric topology* on M. Thus each metric space is naturally a topological space, and all the concepts provided by general topology can be applied. Note that metric topologies are always Hausdorff, since any two points in M can be separated by sufficient small ϵ-balls.

It is easy to define the notion of *boundedness* for arbitrary subsets of a metric space (M, d). For a set $\mathcal{A} \subseteq M$, the nonnegative number

$$d(\mathcal{A}) := \sup\{d(x,y) : x, y \in \mathcal{A}\}$$

is called the *diameter* of $\mathcal{A}$, and $\mathcal{A}$ is said to be *bounded* iff $d(\mathcal{A}) < \infty$.

Boundedness is *not* a topological invariant of the metric topology. For example, consider the real line with the first metric $d_1(x,y) := |x - y|$ and the second metric

$$d_2(x,y) := \frac{|x-y|}{1+|x-y|} \qquad (x, y \in \mathbb{R}).$$

Then the identity mapping $\iota : (\mathbb{R}, d_2) \to (\mathbb{R}, d_1)$ is a homeomorphism, but the diameter of the entire set $\mathbb{R}$ related to the metric d_2 is finite, whereas with respect to the metric d_1 it is infinite.

Let (M_1, d_1) and (M_2, d_2) be metric spaces. Then a mapping $\iota : M_1 \to M_2$ is called an *isometry*, if it satisfies the equations

$$d_2(\iota(x), \iota(y)) = d_1(x,y) \qquad (x, y \in M_1).$$

This definition implies that an isometry is injective. Two metric spaces M_1, M_2 are said to be *isometric* to each other, if there is a *surjective* isometry between them. In particular, isometric spaces are homeomorphic to each other with respect to their metric topologies.

2. Convergence and Completeness. The concept of convergence is already known for sequences in arbitrary topological spaces. By the validness of the first countability axiom in a metric space (M, d), a sequence (x_n) of elements in M converges to a limit point $x \in M$ if and only if $\lim_{n\to\infty} d(x, x_n) = 0$, that is, if for each $\epsilon > 0$ there

is an integer N_ϵ such that $d(x_n, x) < \epsilon$ for all $n > N_\epsilon$. Because each metric space is Hausdorff, the limit point of a convergent sequence must be unique. By the triangle inequality

$$d(x_n, x_m) \leq d(x_n, x) + d(x, x_m),$$

we can see that a convergent sequence in M satisfies the *Cauchy convergence condition*

$$\lim_{n,m\to\infty} d(x_n, x_m) = 0.$$

This condition suggests the following definition: a sequence (x_n) is called *Cauchy sequence*, if for each $\epsilon > 0$ there is a positive integer N_ϵ such that $d(x_m, x_n) < \epsilon$ for all $m, n > N_\epsilon$. In addition, two metrics d_1, d_2 on a set M are called *equivalent*, if they create identical sets of Cauchy sequences, and in this case their corresponding metric topologies coincide.

It is one of the benefits when working with metric spaces that the property of a set to be closed or dense can be expressed in terms of convergent sequences:

1. a subset $\mathcal{A} \subset M$ is closed, iff the limit point of any convergent sequence in $\mathcal{A}$ belongs to $\mathcal{A}$, too,
2. the set $\mathcal{A}$ is dense in M, iff for each point $x \in M$ there is a sequence (x_n) in $\mathcal{A}$ which is convergent to x.

Clearly, every convergent sequence is a Cauchy sequence, but the converse statement does not hold: consider Cauchy sequences in the space $\mathbb{Q}$ of rational numbers, which in general do not have a limit point in $\mathbb{Q}$. This counterexample gives rise to introduce the concept of *completeness* for a metric space, which is of outstanding meaning in higher analysis: indeed, a metric space (M, d) is called *complete*, if each Cauchy sequence (x_n) is convergent to an element $x \in M$.

One can easily prove that the cartesian product of arbitrarily many metric spaces is complete iff each single space is complete, and that a subset $\mathcal{A}$ of a complete metric space is complete if and only if $\mathcal{A}$ is closed.

In the following we shall point out the fundamental assertion that for each *uncomplete* metric space (M, d) there is a *complete* metric space $(\overline{M}, \delta)$ and a continuous embedding $\iota : M \to \overline{M}$, which preserves the metric in the sense that

$$d(x, y) = \delta(\iota(x), \iota(y)) \qquad (x, y \in M).$$

The metric space $\overline{M}$ is called the *completion* of M and is unique up to isometric mappings. Moreover, the image of M is a dense subset in $\overline{M}$.

The space $\overline{M}$ is usually designed as a set of equivalence classes consisting of Cauchy sequences in M. More precisely, two Cauchy sequences (x_n) and (y_n) are called equivalent, if $\lim_{n\to\infty} d(x_n, y_n) = 0$, and the mapping

$$\delta([(x_n)], [(y_n)]) := \lim_{n\to\infty} d(x_n, y_n)$$

defines a metric function on $\overline{M}$, which is independent of the representants in $[(x_n)]$ and $[(y_n)]$. The embedding of M into the metric space $\overline{M}$ is performed by mapping an element $x \in M$ to that equivalent class $[x]$, which contains the *constant* Cauchy sequence $(x, x, \ldots)$. The construction of the real numbers as the completion of the space of rational numbers is a prototypical example for that method.

Complete metric spaces turn out to be Baire spaces.

3. Compactness in Metric Spaces. In the setting of metric spaces the property of compactness and relatively compactness of a given subset can be expressed in terms of *sequentially compactness* (BOLZANO-WEIERSTRASS *theorem*):

THEOREM 3.1. A subset $\mathcal{A} \subseteq M$ of a metric space is relatively compact iff each sequence in $\mathcal{A}$ has a convergent subsequence.

For a compact subset K in M, the limit point of such a subsequence must be a member of K again, since K is closed. Furthermore, every compact metric space is *complete*, due to the fact that each Cauchy sequence in a compact metric space must have a convergent subsequence. In order to describe compact subsets in a metric space, the following concept is adequate: a subset $\mathcal{A} \subseteq M$ is called *totally bounded*, if for each $\epsilon > 0$ there is a finite number of points $x_1, \ldots, x_n \in \mathcal{A}$ such that their ϵ-balls form an open cover of $\mathcal{A}$.

For *complete* metric spaces the *Heine-Borel theorem* asserts that there is a strong link between the concepts of totally boundedness and relatively compactness:

THEOREM 3.2. If a metric space M is complete, then a subset $\mathcal{A} \subset M$ is relatively compact, if and only if it is totally bounded.

For the Euclidean spaces $\mathbb{R}^d$ the notion of totally boundedness and usual boundedness are equivalent, so a subset $K \subset \mathbb{R}^n$ is compact iff K is closed and bounded.

The following property of a compact subset in a metric space is often used: if a set $\mathcal{A}_1$ is *compact* and a further set $\mathcal{A}_2$ is *closed*, then they have non-zero distance to each other, that is, $\delta(\mathcal{A}_1, \mathcal{A}_2) = \delta(\mathcal{A}_2, \mathcal{A}_1) > 0$ holds.

4. Continuity in the Setting of Metric Spaces. In metric spaces there is an alternative approach to define continuity of a mapping: a function $f : D \subseteq M_1 \to M_2$ is called *continuous* at a point $x_0 \in M_1$, if for each $\epsilon > 0$ there exists $\delta > 0$ such that $d_1(x_0, x) < \delta$ implies $d_2(f(x_0), f(x)) < \epsilon$ for all $x \in D$.

Beyond that, the function f is called *continuous*, if f is continuous at every point of its domain D. Notice that in metric spaces a function f is continuous at a point x, iff it is *sequentially continuous* in x, that is, a sequence (x_n) converging to x implies the convergence of the sequence $(f(x_n))$ to the point $f(x)$. The property of a function being continuous at a point x is of local nature.

Let $C(X, \mathbb{C})$ be the set of continuous complex-valued functions $f : X \to \mathbb{C}$. If X is a compact topological space, then any function $f \in C(X, \mathbb{C})$ is bounded. As a consequence, the difference $f - g$ of such functions is bounded, too, and one may introduce a metric d on the set $C(X, \mathbb{C})$ by

$$d(f, g) := \sup_{x \in X} |f(x) - g(x)| \quad \text{for } f, g \in C(X, \mathbb{C}).$$

It is an important task to identify the compact subsets of the set $C(X, \mathbb{C})$ with respect to its metric topology. More precisely, the question arises, under which conditions a subset of $C(X)$ is relatively compact. The main condition in this problem is called *equicontinuity*: a family $\mathfrak{F} \subset C(X)$ is called *equicontinuous*, if for each $\epsilon > 0$ there exists a $\delta > 0$ such that

$$|(f(x) - f(y)| < \epsilon, \text{ whenever } d(x, y) < \delta$$

for every function $f \in \mathfrak{F}$. In addition, the family $\mathfrak{F}$ is called *uniformly bounded* in $x \in X$, if there is a number M such that $|f(x)| \leq M$ for all $f \in \mathfrak{F}$. Once these concepts are available, the answer to the question above is given by the *Arzelá-Ascoli theorem*:

THEOREM 4.1. Let X be a compact space and $C(X)$ the space of continuous complex-valued functions. Then a subset $\mathfrak{F} \subset C(X)$ is relatively compact, iff $\mathfrak{F}$ is equicontinuous and uniformly bounded for each $x \in X$.

There are some generalizations of this theorem. For example, it is still valid for families of continuous mappings $f : X \to M$, where X is a compact Hausdorff space and M is a metric space. Note that in this case the notion of relatively compactness means that for each point $x \in X$ the set $\mathfrak{F}_x := \{f(x) : f \in \mathfrak{F}\}$ is relatively compact.

A further notion in metric space theory is *uniform continuity*, which avoids the dependency of the number δ from the point x: a function $f : D \subseteq M_1 \to M_2$ is called *uniformly continuous*, if for each $\epsilon > 0$ there exists $\delta > 0$, such that $d_1(x, y) < \delta$ implies $d_2(f(x), f(y)) < \epsilon$ for *all* $x, y \in D$. Thus the value δ can be chosen independently of the points x, y.

The HEINE-CANTOR *theorem* quotes:

THEOREM 4.2. If M_1, M_2 are metric spaces and M_1 is *compact*, then every continuous function $f : M_1 \to M_2$ is uniformly continuous.

Uniformly continuous functions map Cauchy sequences to Cauchy sequences again, but this is not true in case of non-uniformly continuous functions. Furthermore, the uniform continuity of a function f is a global property, and each such function clearly is continuous. The converse statement however is not true: consider the mapping $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto x^2$, which is continuous, but *not* uniformly continuous on the real line.

5. Hölder-Continuity. A function $f : M_1 \to M_2$ between metric spaces M_1, M_2 is called *Hölder-continuous*, if there a real constants $c \geq 0$ and $\alpha \geq 0$ such that for all $x, y \in M_1$ the so called *Hölder condition*

$$d_2(f(x), f(y)) \leq c \cdot d_1(x, y)^\alpha$$

is satisfied. Moreover, each Hölder continuous function $f : M_1 \to M_2$ is actually uniformly continuous. In case of $\alpha = 0$, Hölder continuity means that the function f is simply bounded, whereas in case of $\alpha = 1$ it is equivalent to the inequality

$$d_2(f(x), f(y)) \leq c \cdot d_1(x, y),$$

whose validness is known as *Lipschitz continuity.* The number c is called the *Lipschitz constant* of the function f.

If the Lipschitz constant satisfies $0 < c < 1$, the function is called a *contraction.* Contractions in the context of complete metric spaces are very useful, because they ensure the existence and uniqueness of the solution for a couple of nonlinear equations. This fact is based on the Banach *fixpoint theorem*:

Theorem 5.1. Let (M, d) be a complete metric space and the mapping $f : M \to M$ be a contraction. Then f admits exactly one fixpoint of the equation $f(x) = y$.

Indeed, beginning with any $x_0 \in M$, the sequence (x_n) defined by $x_n := f(x_{n-1})$ is Cauchy, whose limit point x is the solution of the equation $f(x) = x$.

III. Topological Groups

Besides the metric spaces there is a further subclass of topological spaces, which is of great importance in analysis, namely that of *topological groups.*

First recall from group theory that an arbitrary group is a set G together with an associative operation $\circ : G \times G \to G$, such that

1. there is exactly one *identity element* $1 \in G$ such that
$$x \circ 1 = 1 \circ x = x \qquad (x \in G),$$
2. to each $x \in G$ there is a unique element $x^{-1} \in G$, called the *inverse* of x such that
$$x \circ x^{-1} = x^{-1} \circ x = 1.$$

If the group operation is commutative, then the identity element is often denoted 0 and each inverse element x^{-1} is denoted $-x$.

1. Generalities on Topological Groups. These are mathematical objects having the algebraic structure of a group and a topological structure together. Both kinds must be compatible to each other in the following sense: let $(G, \circ)$ be an ordinary group and a topological space $(G, \mathcal{T})$ at once. Then G is called a *topological group*, if the group operation
$$\circ : G \times G \to G, \quad (x, y) \mapsto x \circ y^{-1}$$

is a continuous mapping. Note that instead to require continuity for the mapping $(x, y) \mapsto x \circ y^{-1}$, one can equivalently require that the mappings $(x, y) \mapsto x \circ y$ and $x \mapsto x^{-1}$ be continuous.

Well-known examples of topological groups are:

1. each group G with its discrete topology 2^G,
2. the group $\mathbb{R}^\times := \mathbb{R} - \{0\}$,
3. the group $\mathbb{C}^\times := \mathbb{C} - \{(0,0)\}$,
4. the group $\mathbb{R}^+ := \{\, x \in \mathbb{R} \,:\, x > 0 \,\}$,
5. the additive groups $(\mathbb{R}, +)$ and $(\mathbb{C}, +)$,
6. each subgroup U of a topological group G endowed with the subspace topology and its closure $\overline{U}$ are topological groups in its own right.

Like for purely algebraic groups, a *homomorphism* between topological groups is defined to be a *continuous* group homomorphism, and a homomorphism is called an *isomorphism*, if it is continuous and a group isomorphism at the same time. A well-known example of a group isomorphism is the *exponential function* $t \mapsto e^t$, which maps the topological group $(\mathbb{R}, +)$ to the group $(\mathbb{R}^+, \cdot)$. Similarly, the function

$$t \in [0, 1) \mapsto e^{2\pi i t}$$

forms a group isomorphism between the groups $\mathbb{R}/\mathbb{Z}$ and $\mathbb{T}$.

Since in a topological group the left translation $x \mapsto g \circ x$ and the right translation $x \mapsto x \circ g$ are continuous mappings, they are both examples for *automorphisms*, that is, isomorphisms on the group G to itself. Hence the topological properties of a topological group G are completely determined by the topology in a neighbourhood of the identity element $1 \in G$, which gives rise to call a topological group to be *homogeneous*.

The kind, how the identity 1 of a topological group G lies in a given subgroup U, determines the topology of that subgroup. In particular,

1. $U \subset G$ is open, iff 1 is an inner point of U.
2. U is discrete, iff 1 is an isolated point of U.

The *circle group* $\mathbb{T}$ is that group consisting of all complex numbers of modulus 1 with group operation

$$e^{i\alpha} \circ e^{i\beta} := e^{i\alpha} \cdot e^{i\beta} := e^{i(\alpha+\beta)}, \quad \alpha, \beta \in \mathbb{R}.$$

This group, endowed with the subset topology inherited by the standard topology on the field $\mathbb{C}$, is a topological group. Consider the set $C(G, \mathbb{T})$ consisting of all continuous functions $f : G \to \mathbb{T}$. We equip this set with the compact-open topology, so $C(G, \mathbb{T})$ becomes into a Hausdorff space.

2. Characters and Dual Groups. A *character* is a group homomorphism from a topological group G to the circle group $\mathbb{T}$ (in the sense of topological groups) and hence a continuous mapping. We denote G^* the set of all characters $\chi : G \to \mathbb{T}$, which turns out to be a subset of the function space $C(G, \mathbb{T})$. The set G^* can be made into a topological space by choosing the topology $\mathcal{T}_{G^*}$ on G^* to be the subset topology with respect to the compact-open topology on $C(G, \mathbb{T})$.

Now, performing multiplication of characters $\chi_1, \chi_2 \in G^*$ by

$$\chi_1 \circ \chi_2 : g \mapsto \chi_1(g) \cdot \chi_2(g),$$

we obtain a further character $\chi_1 \circ \chi_2 \in G^*$. This operation has all the properties of a group operation, and since it is commutative, the set G^* of all characters of G has the structure of an Abelian group, called the *character group* or the *dual group* of G.

Furthermore, one can prove that the group operation in G^* is continuous with respect to the topology $\mathcal{T}_{G^*}$, hence the dual group of any topological group is a topological group as well.

In the following table we list three classical topological Abelian groups together with their dual groups and characters:

G	G^*	characters
$\mathbb{R}$	$\mathbb{R}^* \cong \mathbb{R}$	$\chi_x(t) := e^{itx}$, $x \in \mathbb{R}$
$\mathbb{Z}$	$\mathbb{Z}^* \cong \mathbb{T}$	$\chi_z(n) := e^{inz}$, $z \in \mathbb{T}$
$\mathbb{T}$	$\mathbb{T}^* \cong \mathbb{Z}$	$\chi_n(z) := z^n$, $n \in \mathbb{Z}$

Table 1: classical topological groups and dual groups.

Concerning cartesian products of topological groups, for each topological group G and H we have

$$(G \times H)^* \cong G^* \times H^*.$$

As a consequence, the group isomorphisms $(\mathbb{R}^d)^* \cong \mathbb{R}^d$ hold for any integer d.

3. Pontryagin Duality. The abbreviation *LCA group* usually stands for a *locally compact Abelian group*. The meaning of locally compact groups is based on the fact that they own non-trivial characters, that is, we have $G^* \neq \{\chi_1\}$ for such a group G, where χ_1 is the character with $\chi_1(x) = 1$ for all $x \in G$.

Suppose that the group G is *locally compact*. Then it can be shown (for example by the ARZELÁ-ASCOLI theorem) that the dual group G^* is also locally compact with respect to its compact-open topology. Let G^{**} denote the *bidual group* of a topological group G, which is defined to be the dual group of its character group.

One of the main results in the theory of LCA groups is the celebrated *Pontryagin duality theorem*:

THEOREM 3.1. If G is a LCA group, then G and its bidual group G^{**} are isomorphic to each other.

Notice that each group element $g \in G$ gives rise to a character χ_x on the dual group G^* defined by $\chi_x(\xi) := \xi(x)$ for all $\xi \in G^*$. Thus the mapping $x \mapsto \chi_x \in G^{**}$ is an injective group homomorphism, and Pontryagin duality means that the canonical embedding $\iota : G \to G^{**}$, $x \mapsto \chi_x$ is also surjective.

It is easy to see that the groups listed in table 1 are examples for Pontryagin duality. This kind of duality also implies remarkable relationships between the topological properties of the LCA group G and its dual group G^*:

1. G is compact if and only if G^* is discrete.
2. G is discrete if and only if G^* is compact.

References.

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